

5.1 SIMPLIFICATIONS

In regular routine of our life, we add, multiply, subtract or divide two numbers for different purposes. After each of those cases we get our result as a simplified number. The process of getting a simplified number is known as **simplification**.

Consider this expression.

$$3 + 4 \times 2$$

We can simplify it by two ways:

- (i) If we perform addition of 3 and 4 and then multiply the sum by 2, we get the answer 14.
- (ii) If we perform multiplication of 4 and 2 and then add 3 to the product 8, we get the answer 11.

So to get a unique result of our simplification process, we must follow some **rules of preference**; we use brackets to show the preference of operations to be performed.

To know the Kinds of Brackets

We know that for simplification brackets are used to group two or more numbers together with operations.

There are four kinds of brackets.

- (i) — Vinculum or bar
- (ii) () parenthesis or curved brackets or round brackets
- (iii) { } braces or curly brackets
- (iv) [] square brackets

Know the order of preference as, --- , $()$, $\{ \}$ and $[]$, to remove (simplify) them from an expression.

In problems involving more than one bracket, the brackets should be removed in the order --- , $()$, $\{ \}$ and $[]$. Removing brackets means simplifying expression within the brackets to get the simplified form and then remove the pair of brackets.

We use following rules to remove the brackets.

- (i) If there is a plus '+' sign before brackets, the brackets are removed without changing the sign of the number within brackets.

For example, $+ (2 - 5) = + (-3) = -3$

- (ii) If there is a minus '-' sign before brackets, the brackets are removed and the sign of the number within brackets is changed.

For example, $-(2 - 5) = -(-3) = 3$

- (iii) If there is a number before brackets, the number in the brackets left after simplification is multiplied by this number and the brackets are removed.

For example, $4(2 - 5) = 4(-3) = -12$

Also $-5(3 - 7) = -5(-4) = +20$

Example 1: Simplify : $5 + \{16 - 4 \div 2 \times 3 - (-6 \div 2)\}$

Solution: $5 + \{16 - 4 \div 2 \times 3 - (-6 \div 2)\}$

$$= 5 + \{16 - 2 \times 3 - (-3)\}$$

$$= 5 + \{16 - 6 + 3\}$$

$$= 5 + (10 + 3)$$

$$= 5 + 13 = 18$$

Example 2: Simplify : $2 [-2 \{4 \times 9 \div 8 (5 - \overline{3 - 4})\}]$

$$\begin{aligned}
 \text{Solution: } & 2 [-2 \{4 \times 9 \div 8 (5 - \overline{3 - 4})\}] \\
 &= 2 [-2 \{4 \times 9 \div 8 (5 - (-1))\}] \\
 &= 2 [-2 \{4 \times 9 \div 8 (5 + 1)\}] \\
 &= 2 [-2 \{4 \times 9 \div 8 \times 6\}] \\
 &= 2 [-2 [4 \times \frac{9}{8} \times 6]] \\
 &= 2 [-2 \{\frac{\overset{1}{\cancel{4}} \times 9 \times \overset{3}{\cancel{6}}}{\underset{2}{\cancel{8}}}\}] \\
 &= 2 [-2 \{9 \times 3\}] \\
 &= 2 [-2 \times 27] \\
 &= 2 \times -54 \\
 &= -108
 \end{aligned}$$

5.2 BODMAS RULE

Recognize BODMAS rule to follow the order in which the operations, to simplify mathematical expressions, are performed.

BODMAS indicates the sequence in which more than one operations are performed in the simplification process of any mathematical problem.

The chain of letters in BODMAS shows meaning of each letter as:

- B for "Brackets"
- O for "order of operation"
- D for "Division"
- M for "Multiplication"
- A for "Addition"
- S for "Subtraction"

Removal of brackets and solving of operation 'of' with operations of DMAS is known as BODMAS rule.

In BODMAS rule, the **of** denotes order of operation.

Then we perform the operations of division, multiplication, addition and subtraction (DMAS) in sequence from left to right. By using BODMAS rule we can simplify any expression involving brackets and arithmetical operations to simplified form.

In short, to follow the BODMAS rule, we first remove **brackets**. Next we perform the operation '**of**'. Then the operation of **division** is performed followed by the operation of **multiplication**. Lastly we perform **addition** followed by the operation of **subtraction**.

After learning about the sequence of operations, we can now apply them to complete the simplification process.

Simplify mathematical expressions involving fractions and decimals grouped with brackets using BODMAS rule

Let us solve some examples using BODMAS rule.

Example 1. Simplify $3\frac{1}{4} \div \left[\frac{2}{5} + \left\{ \left(\frac{7}{8} - \frac{7}{12} \right) + 1\frac{5}{6} \right\} - 2\frac{2}{5} \right]$

$$\begin{aligned}
 \text{Solution: } & 3\frac{1}{4} \div \left[\frac{2}{5} + \left\{ \left(\frac{7}{8} - \frac{7}{12} \right) + 1\frac{5}{6} \right\} - 2\frac{2}{5} \right] \\
 &= \frac{13}{4} \div \left[\frac{2}{5} + \left\{ \left(\frac{7}{8} - \frac{7}{12} \right) + \frac{11}{6} \right\} - \frac{12}{5} \right] \quad \text{(Converting mixed fractions into improper fractions)} \\
 &= \frac{13}{4} \div \left[\frac{2}{5} + \left\{ \left(\frac{21 - 14}{24} \right) + \frac{11}{6} \right\} - \frac{12}{5} \right] \quad \text{(Simplifying within parenthesis)} \\
 &= \frac{13}{4} \div \left[\frac{2}{5} + \left\{ \frac{7}{24} + \frac{11}{6} \right\} - \frac{12}{5} \right] \quad \text{(Removing parenthesis)} \\
 &= \frac{13}{4} \div \left[\frac{2}{5} + \left\{ \frac{7 + 44}{24} \right\} - \frac{12}{5} \right] \quad \text{(Simplifying within braces)}
 \end{aligned}$$

$$= \frac{13}{4} \div \left[\frac{2}{5} + \frac{51}{24} - \frac{12}{5} \right] \quad (\text{Removing braces})$$

$$= \frac{13}{4} \div \left[\frac{48 + 255 - 288}{120} \right] \quad (\text{Simplifying within square brackets})$$

$$= \frac{13}{4} \div \frac{15}{120} \quad (\text{Removing square brackets})$$

$$= \frac{13}{\cancel{4}_1} \times \frac{\overset{2}{\cancel{120}^{30}}}{\cancel{15}_1} = 13 \times 2$$

$$= 26$$

Example 2. Simplify: $1\frac{3}{4} \div \left[9\frac{5}{8} \div \left\{ \frac{2}{5} + \left(8\frac{2}{5} \times 4\frac{1}{6} - 3\frac{1}{2} \right) \right\} \right]$

Solution: $1\frac{3}{4} \div \left[9\frac{5}{8} \div \left\{ \frac{2}{5} + \left(8\frac{2}{5} \times 4\frac{1}{6} - 3\frac{1}{2} \right) \right\} \right]$

$$= \frac{7}{4} \div \left[\frac{77}{8} \div \left\{ \frac{2}{5} + \left(\frac{42}{5} \times \frac{25}{6} - \frac{7}{2} \right) \right\} \right] \quad (\text{Converting mixed fractions into improper fractions})$$

$$= \frac{7}{4} \div \left[\frac{77}{8} \div \left\{ \frac{2}{5} + \left(\frac{42}{5} \times \frac{25-21}{6} \right) \right\} \right] \quad (\text{Solving vinculum})$$

$$= \frac{7}{4} \div \left[\frac{77}{8} \div \left\{ \frac{2}{5} + \left(\frac{\cancel{42}}{5} \times \frac{4}{\cancel{6}_1} \right) \right\} \right] \quad (\text{Solving within parenthesis})$$

$$= \frac{7}{4} \div \left[\frac{77}{8} \div \left\{ \frac{2}{5} + \frac{28}{5} \right\} \right] \quad (\text{Removing parenthesis})$$

$$= \frac{7}{4} \div \left[\frac{77}{8} \div \left\{ \frac{\overset{6}{\cancel{30}}}{\cancel{5}_1} \right\} \right] \quad (\text{Solving braces})$$

$$= \frac{7}{4} \div \left[\frac{77}{8} \times \frac{1}{6} \right] \quad (\text{Removing braces})$$

$$= \frac{7}{4} \div \frac{77}{48} = \frac{7}{4} \times \frac{48}{77} = \frac{12}{11} = 1\frac{1}{11}$$

Example 3. Simplify: $6.4 - \overline{(3.3 - 1.2 \times 2.5)}$

Solution: $6.4 - \overline{(3.3 - 1.2 \times 2.5)}$ (Solving viculum)

$$= 6.4 - (2.1 \times 2.5) \quad (\text{Solving within parenthesis})$$

$$= 6.4 - 5.25 \quad (\text{Removing parenthesis})$$

$$= 1.15$$

Example 4. Simplify: $2.9 \times \overline{1.8 - 1.3} \times 3.75$

Solution: $2.9 \times \overline{1.8 - 1.3} \times 3.75$

$$= 2.9 \times 0.5 \times 3.75$$

$$= 5.4375$$

Example 5. Simplify: $3.05 + \left[2.34 \div \left\{ 2.4 - \overline{(1.8 \times 0.3 + 0.6)} \right\} \right]$

Solution: $3.05 + \left[2.34 \div \left\{ 2.4 - \overline{(1.8 \times 0.3 + 0.6)} \right\} \right]$

$$= 3.05 + \left[2.34 \div \left\{ 2.4 - (1.8 \times 0.9) \right\} \right]$$

$$= 3.05 + \left[2.34 \div \left\{ 2.4 - 1.62 \right\} \right]$$

$$= 3.05 + \left[2.34 \div 0.78 \right]$$

$$= 3.05 + 3$$

$$= 6.05$$

EXERCISE 5.1

Simplify the following:

$$1. \quad \frac{1}{5} - \left\{ \frac{2}{5} \div \frac{4}{15} \times \frac{1}{2} - \left(\frac{1}{2} + \frac{2}{5} \right) \right\}$$

$$2. \quad \frac{3}{4} + \left(\frac{4}{3} - \frac{1}{2} + \frac{2}{3} + \frac{4}{5} \right)$$

$$3. \quad \frac{7}{8} + \left\{ \frac{5}{7} + \left(\frac{1}{2} + \frac{1}{3} \right) \div \frac{5}{6} \right\}$$

$$4. \quad \frac{5}{2} \times \left(\frac{5}{6} - \frac{2}{3} \right) \div \frac{2}{5} \times \left(\frac{3}{5} + \frac{6}{25} \right)$$

$$5. \quad \left(\frac{2}{3} - \frac{3}{4} \times \frac{1}{8} \right) \div \left(\frac{2}{3} - \frac{3}{4} + \frac{5}{8} \right)$$

$$6. \quad 1\frac{1}{2} + \left[\frac{1}{3} \div \left\{ \frac{2}{5} \div \left(\frac{4}{5} - \frac{1}{10} + \frac{3}{5} \right) \right\} \right]$$

$$7. \quad \left[\frac{1}{2} + \left\{ \left(\frac{3}{4} \div \frac{9}{16} \right) \times 1\frac{1}{2} \right\} - 2\frac{1}{4} \right]$$

$$8. \quad 3\frac{1}{2} + \left\{ \left(10\frac{2}{5} - 5\frac{1}{3} \right) \div 3\frac{2}{3} \right\} - 1\frac{1}{5}$$

$$9. \quad \left[\left(2\frac{3}{4} + 1\frac{2}{6} \right) \times \left(1\frac{1}{5} \div \frac{2}{5} \right) \right]$$

$$10. \quad \left\{ \left(\frac{1}{2} - \frac{1}{3} \right) + \left(1\frac{2}{3} \div \frac{1}{9} \right) \right\} - \left\{ \left(\frac{5}{3} - \frac{1}{6} \right) \div \frac{9}{6} \right\}$$

$$11. \quad \left[2\frac{1}{4} \times \left\{ 3\frac{1}{3} - \frac{1}{2} + \left(\frac{5}{3} - \frac{1}{6} \right) \right\} \right] + \frac{5}{6}$$

$$12. \left(3\frac{1}{2} \times 5\frac{1}{3} - 10\frac{2}{5}\right) \times \left(3\frac{1}{8} - 6\frac{1}{2} \div 2\frac{1}{6}\right)$$

$$13. \left(2\frac{1}{5} \div 3\frac{2}{25}\right) \times 4\frac{2}{5} \div \left(3 - \frac{1}{5}\right)$$

$$14. 4.02 \times \left\{1.2 \times \overline{2.11 + 3.89}\right\}$$

$$15. 4.093 + \left\{5.2 \times \overline{1.5 + 2.4}\right\}$$

$$16. 5.7 \times \left\{6.7 + \left(1.3 \times 7.2 \div \overline{1.5 \times 2.4}\right)\right\}$$

$$17. 0.3 \times 2.6 + \left\{3.7 - \left(1.2 \times \overline{4.5 \div 1.5}\right)\right\}$$

$$18. \left\{\left(3.41 - \overline{1.05 + 1.5}\right) \times 3.5\right\} \div 1.25$$

$$19. 30.45 + \left[6.35 \times \left\{8.95 - \left(3.5 \div 0.7 \times 0.01\right)\right\}\right]$$

$$20. 2.297 + \left[0.2 \times \left\{9 \div 0.3\right\} - 0.09\right]$$

$$21. \left\{\left(2.5 - 2\right) + 1.5\right\} \times 0.5 - 0.05$$

$$22. 2.04 + \left[2.82 \div \left\{2.4 - \left(1.3 \times 0.3 + 0.6\right)\right\}\right]$$

$$23. \left[\left\{7.45 - 1.35 + 1.5\right\} \times 1.25 \div 2.5\right]$$

Solve real life problems involving fractions and decimals

Fractions and decimals are used in daily life as explained in the following examples.

Example 1.

The maximum marks of Urdu test are 40 and passing marks are $\frac{1}{4}$ of maximum marks. How many marks at least should Haris obtain to pass the test?

$$\begin{aligned}\text{Solution: Maximum marks} &= 40 \\ \text{Minimum passing marks} &= \frac{1}{4} \text{ of } 40 \\ &= \frac{1}{4} \times 40 \\ &= 10\end{aligned}$$

Therefore, Haris should obtain at least **10 marks** to pass the test.

Example 2.

Fouzia gave $\frac{1}{4}$ of her cake to her sister, another $\frac{2}{3}$ to her two brothers and kept the rest of the cake for herself. What fraction of the cake was left with her.

Solution: Fouzia has 1 cake
She gave $\frac{1}{4}$ and $\frac{2}{3}$ of the cake to her sister and two brothers respectively.

$$\begin{aligned}\text{Total cake given} &= \frac{1}{4} + \frac{2}{3} \\ \text{Cake left with Fouzia} &= 1 - \left(\frac{1}{4} + \frac{2}{3}\right) = 1 - \left(\frac{3+8}{12}\right) \\ &= 1 - \frac{11}{12} = \frac{12-11}{12} = \frac{1}{12}\end{aligned}$$

Hence $\frac{1}{12}$ th part of the cake was left with Fouzia.

Example 3. There are 5380 suits in a shop. $\frac{3}{5}$ of these are in red colour and $\frac{1}{10}$ these are in green colour and the remaining suits are in multi colours. Find the number of suits in multi colours.

Solution:

$$\text{The number of suits in red colour} = \frac{3}{5} \times \frac{1076}{5350} = 3228$$

$$\text{The number of suits in green colour} = \frac{1}{10} \times 5350 = 538$$

$$\begin{aligned} \text{The number of suits in multi colours} &= 5380 - (3228 + 538) \\ &= 5380 - 3766 \\ &= 1614 \end{aligned}$$

Hence the number of suits in multi colours are 1614.

Example 4.

Najeeb's father bought the following items from market:

- (i) 30 eggs at the rate of Rs 10.50 per egg.
- (ii) 20 bars of washing soap at the rate of Rs 52.75 per bar.
- (iii) 25 pieces of toilet soap at the rate of Rs 45.50 per piece.
- (iv) 15 packets washing powder at the rate of Rs 120.80 per packet.

How much money is paid by him?

Solution:

Cost of eggs	=	30	×	10.50	=	Rs	315.00
Cost of washing soap	=	20	×	52.75	=	Rs	1055.00
Cost of toilet soap	=	25	×	45.80	=	Rs	1145.00
Cost of washing powder	=	15	×	120.80	=	+ Rs	1812.00
Total							Rs 4327.00

Total amount spent = 315 + 1055 + 1145 + 1812 = 4327

Hence Najeeb's father paid Rs 4327.

EXERCISE 5.2

1. There are 640 students in total in a school. $\frac{3}{10}$ of the students use own vehicles, $\frac{5}{8}$ use public transport and the remaining students come on foot. How many students of the school come on foot?
2. From a box of 140 oranges, $\frac{3}{7}$ of the oranges were sold and $\frac{1}{14}$ got spoilt. What is the number of the oranges left in the box?
3. A crowd of people was gathered for a cricket match. $\frac{1}{8}$ of crowd had come by train, $\frac{2}{7}$ by bus, $\frac{1}{4}$ by own vehicles and the remaining people came on foot. Find the part of people who came on foot.
4. Maira had Rs 58,000. She gave $\frac{16}{29}$ of it to her mother and $\frac{1}{4}$ of the remainder in Zakat fund. Find the amount left with her?
5. Aamir planted 425 plants of tomatoes. After some days 25 plants died. What fraction of plants was survived?
6. Aijaz gave $\frac{2}{5}$ of his money to his daughter, $\frac{1}{7}$ to his son and the remainder to his wife. What fraction of the amount was given to his wife.

7. Adeel had 26.50 metres of cloth. He used 5.80 metres of cloth for his suit. He gives 11.50 metres of cloth to his brother. How much cloth is left with Adeel in fraction?
8. Anum's family purchased 5 packets of cooking oil at the rate of Rs 121.80 per packet, 3 bags of rice at the rate of Rs 235.50 per bag and 40 kg flour at the rate of Rs 42.70 per kg. Find the total amount paid by her.
9. Umair spent 0.06 of his pocket money on Monday, 0.25 on Tuesday and 0.50 on Wednesday. Remaining money spent on Thursday. What fraction of his pocket money he had on Thursday?
10. Sobia purchased the following items for her home.
 - (i) Two electric bulbs at the rate of Rs 80.95 each.
 - (ii) Four small stools at the rate of Rs 105.50 each.
 - (iii) Three tables at the rate of Rs 530.95 each.
 - (iv) Six chairs at the rate of Rs 458.30 each.

Find the total amount paid by her.

REVIEW EXERCISE 5

1. Simplify: (i) $1\frac{2}{5} \div \left\{ 2\frac{3}{5} - \left(1\frac{1}{4} - \frac{2}{3} \times 1\frac{1}{7} \right) \right\}$
- (ii) $2\frac{1}{4} \div \left[\frac{2}{3} \div \left\{ \frac{1}{3} - \left(\frac{1}{2} - \frac{1}{5} \div 1\frac{1}{2} \right) \right\} \right]$
- (iii) $4.3 \times [3.9 \times 0.8 - 1.3 + \{4.8 \div 1.2 - (1.3 - 1.2 \times 2.5)\}]$
- (iv) $2.9 - [1.05 \times 1.17 + (1.04 - 2.5 + 1.9)]$

2. Solve the following:
- (i) Bilal sacrifices a goat and get $13\frac{5}{7}$ kg of meat. He gave one third of meat in charity. How much meat is left with him?
 - (ii) Azhar solved 15 questions out of 25 questions of mathematics. Find the fractions of solved and unsolved questions.
3. Fareena read $\frac{2}{5}$ of the book in one day. Remaining of the book was read on next day. What part of the book was read on next day? If book contains 150 pages, then find the number of pages she read on first day and the next day.
4. A contractor purchased tiles of different qualities for a building repair as under:
- (i) Local Tiles 240.34 square metres at the rate of Rs 327.34 per square metre.
 - (ii) Imported Tiles 105.28 square metres at the rate of Rs 539.99 per square metre.
- Find the total amount paid by the contractor.
5. Write true (T) or false (F) for the following:
- (i) The symbol for vinculum is []
 - (ii) In simplification, if there is a negative sign before a bracket, then the sign inside the bracket remains unchanged, when removing the brackets.
 - (iii) The expression $35 + 2 \times 8$ is simplified by first multiplying 2 by 8 and then adding 35 to it.
6. Fill in the blanks:
- (i) In BODMAS, M stands for _____.
 - (ii) The bracket { } is called _____.
 - (iii) Brackets are simplified in the order as: vinculum, _____, _____ and _____.

SUMMARY

- BODMAS indicates the sequence in which more than one operations are performed in the simplification process of any mathematical expression.
Where, **B** stands for **bracket**, **O** for the order of operation **of**, **D** for **division**, **M** for **multiplication**, **A** for **addition** and **S** for **subtraction**.
- While simplifying mathematical expressions involving brackets should be removed in the order first —, second () then { } and [].
- The operations, **division**, **multiplication**, **addition**, and **subtraction** (DMAS) are performed from left to right in order sequence.
- If there is a **plus sign before brackets**, the **brackets** are removed **without changing** the sign of the number within the brackets.
- If there is a **minus sign** before a **pair of brackets**, the sign of the number within the **second brackets is changed** and brackets are removed.
- If there is a number before brackets, the number in the brackets left after simplification is multiplied by this number and the brackets are removed.